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SOE-BVAR Suite — Model Scorecard

Data source: real (RBA + ABS + FRED)

Panel: 1997Q4-2026Q1 | **Targets:** gdp_growth, cpi_inflation, unemp_rate, cash_rate | **Horizons:** 1-12 quarters

Evaluation: expanding-window pseudo-real-time, 36 forecast origins; densities scored by CRPS and log predictive density, points by RMSE, all by horizon.

Headline metric — the LEVEL error. Performance is reported on the forecast error of each series' *level* at quarter $t+h$: for GDP and inflation (modelled as growth) the cumulative level from the forecast origin — where real GDP and the price level land — and for unemployment and the cash rate the rate level itself. The quarterly and year-ended *growth* scores are not shown here (they answer a different, narrower question); they remain in `output/tables/scores_by_horizon.csv`.

Diagnostics (§9): all green (block exogeneity, MCMC convergence, forecast sanity, no-look-ahead, reproducibility).

1. The models

Every VAR member is **block-exogenous** (the domestic block never feeds back into the foreign block) and produces **iterated** density forecasts. Members are designed to fail differently; see README.md for the full rationale.

1a. VAR members

Model	Family	System	Lags	Shrinkage λ	Volatility	COVID
<code>small_minn</code>	Independent Normal-inverse- Wishart (Gibbs) + SOC/DIO	small (8 var)	4	auto (GLP)	constant	LP scaling
<code>small_ss</code>	Steady-state (Vil- lani)	small (8 var)	4	auto (GLP)	constant	LP scaling
<code>small_sv</code>	Stochastic volatility, equa- tion-by-equation	small (8 var)	4	auto (GLP)	stochastic (SV)	t-errors (SV-t)
<code>small_loose_p5</code>	Independent Normal-inverse- Wishart (Gibbs)	small (8 var)	5	0.4	constant	LP scaling
<code>medium_minn</code>	Stochastic volatility, equa- tion-by-equation	medium (13 var)	2	auto (GLP)	stochastic (SV)	t-errors (SV-t)
<code>medium_conj</code>	Block-recursive conjugate NIW + SOC/DIO	medium (13 var)	4	0.1	constant	LP scaling
<code>small_tight</code>	Block-recursive conjugate NIW + SOC/DIO	small (8 var)	4	0.05	constant	LP scaling

1b. Benchmark members (the bar every VAR must clear)

Model	Description	COVID
<code>rw</code>	Random walk	LP scaling
<code>ar4</code>	Bayesian AR(4)	LP scaling

Model	Description	COVID
ucsv	Unobserved components + stochastic volatility	t-errors (robust)
ucmean	Unconditional mean	LP scaling

1c. Combination schemes (density pools)

Weights estimated **per target variable and per horizon bucket** (near 1-4, medium 5-8, far 9-12), shrunk toward equal weights, strictly recursive (no look-ahead).

Scheme	How weights are set
combo_equal	Equal weights (the benchmark pool — hard to beat)
combo_logscore	Recursive log-score weights, with forgetting
combo_pool	Optimal prediction pool (Hall-Mitchell / Geweke-Amisano)
combo_bma	Bayesian model averaging — reported as a diagnostic only

2. Forecast performance — level errors

All tables below score the **level** of each series at $t+h$ (§ definition in the header): cumulative real GDP and the price level for the growth variables, the rate level for unemployment and the cash rate. Lower CRPS / RMSE is better; higher log score is better. **Bold** = best in that column. Models ordered best-first (mean over the shown horizons). The level errors grow with horizon (they accumulate the whole path), so the columns are not comparable across horizons — read down each column, not across.

2a. Who forecasts the level best, by variable and horizon bucket (CRPS)

Best single model (lowest mean level CRPS in the bucket; value in parentheses):

Variable	near (1-4)	medium (5-8)	far (9-12)
gdp_growth	medium_minn (0.988)	small_sv (1.693)	small_sv (2.156)
cpi_inflation	rw (0.386)	rw (1.358)	ar4 (2.693)
unemp_rate	medium_minn (0.318)	small_ss (0.631)	small_ss (0.721)
cash_rate	small_sv (0.307)	combo_equal (0.893)	ar4 (1.447)

2b. Level density accuracy by variable and horizon (CRPS, lower better)

Real GDP growth (qtr %) ([gdp_growth](#)) — cumulative level from the origin

Model	h=1	h=4	h=8	h=12
small_sv	0.611	1.267	1.954	2.206
medium_minn	0.610	1.275	1.995	2.314
combo_equal	0.634	1.305	2.030	2.237
small_tight	0.682	1.366	2.101	2.165
ucsv	0.634	1.304	2.020	2.424
medium_conj	0.672	1.383	2.139	2.238
small_ss	0.664	1.377	2.169	2.240
small_loose_p5	0.627	1.376	2.173	2.313
ar4	0.668	1.342	2.156	2.470
small_minn	0.674	1.420	2.259	2.337
ucmean	0.680	1.380	2.273	2.662
combo_pool	0.642	1.425	2.651	3.685

Model	h=1	h=4	h=8	h=12
combo_logscore	0.754	1.789	4.352	3.995
combo_bma	0.842	2.203	6.319	5.314
rw	0.941	3.264	6.711	11.084

Trimmed-mean CPI inflation (qtr %) ([cpi_inflation](#)) – cumulative level from the origin

Model	h=1	h=4	h=8	h=12
rw	0.131	0.674	1.832	3.356
ar4	0.162	0.844	1.973	3.108
combo_equal	0.161	0.770	1.919	3.340
combo_pool	0.147	0.761	2.043	3.523
combo_logscore	0.137	0.729	2.075	3.592
small_minn	0.170	0.827	2.113	3.714
small_ss	0.172	0.849	2.171	3.663
small_sv	0.170	0.849	2.129	3.761
ucmean	0.230	1.056	2.269	3.405
medium_minn	0.165	0.814	2.141	4.002
combo_bma	0.145	0.800	2.363	3.936
small_tight	0.198	0.925	2.241	3.910
medium_conj	0.191	0.895	2.234	4.029
ucsv	0.175	0.920	2.331	4.056
small_loose_p5	0.192	0.935	2.436	4.117

Unemployment rate (%) ([unemp_rate](#)) – rate level at t+h

Model	h=1	h=4	h=8	h=12
small_ss	0.146	0.500	0.688	0.747
medium_minn	0.146	0.466	0.722	0.796
combo_equal	0.159	0.490	0.699	0.782
small_sv	0.154	0.501	0.758	0.782
small_minn	0.152	0.529	0.753	0.787
small_loose_p5	0.155	0.546	0.780	0.808
small_tight	0.158	0.549	0.779	0.811
combo_pool	0.174	0.490	0.765	0.871
ucsv	0.265	0.531	0.737	0.832
combo_logscore	0.232	0.532	0.774	0.848
rw	0.169	0.545	0.777	0.913
medium_conj	0.163	0.545	0.820	0.896
ar4	0.189	0.534	0.760	0.955
combo_bma	0.148	0.475	0.935	0.886
ucmean	0.726	0.774	0.905	1.034

Cash rate (%) (**cash_rate**) – rate level at t+h

Model	h=1	h=4	h=8	h=12
combo_equal	0.147	0.514	1.126	1.668
ar4	0.167	0.589	1.141	1.593
small_ss	0.139	0.525	1.151	1.676
rw	0.183	0.590	1.136	1.670
small_minn	0.145	0.527	1.152	1.785
combo_pool	0.130	0.547	1.223	1.724
combo_logscore	0.122	0.532	1.223	1.758
ucsv	0.170	0.639	1.266	1.832
combo_bma	0.124	0.576	1.393	1.850
small_tight	0.157	0.598	1.286	1.935
medium_conj	0.192	0.572	1.239	2.053
small_loose_p5	0.162	0.609	1.276	2.076
small_sv	0.115	0.533	1.422	2.111
medium_minn	0.113	0.559	1.462	2.467
ucmean	1.563	1.654	1.671	1.615

2c. Level point accuracy by variable and horizon (RMSE, lower better)

gdp_growth

Model	h=1	h=4	h=8	h=12
small_sv	1.583	2.554	3.713	3.739
small_tight	1.614	2.623	3.654	3.830
small_ss	1.603	2.599	3.682	3.845
ucsv	1.599	2.556	3.620	4.045
small_minn	1.612	2.636	3.735	3.938
ar4	1.632	2.632	3.631	4.036
small_loose_p5	1.566	2.608	3.691	4.079
ucmean	1.619	2.634	3.685	4.152
medium_conj	1.616	2.690	3.774	4.030
medium_minn	1.578	2.644	3.803	4.319
combo_equal	1.620	2.818	4.130	4.938
combo_pool	1.688	3.361	6.337	8.822
combo_logscore	2.054	5.603	11.629	8.818
combo_bma	2.306	6.971	15.316	11.126
rw	2.377	7.807	15.379	24.092

cpi_inflation

Model	h=1	h=4	h=8	h=12
ar4	0.315	1.433	2.905	4.138

Model	h=1	h=4	h=8	h=12
ucmean	0.434	1.648	3.084	4.269
combo_equal	0.326	1.501	3.322	5.216
combo_pool	0.285	1.443	3.361	5.454
combo_logscore	0.261	1.388	3.452	5.668
small_ss	0.332	1.589	3.549	5.427
small_minn	0.333	1.587	3.543	5.511
rw	0.235	1.301	3.486	6.042
small_tight	0.402	1.736	3.655	5.611
small_sv	0.357	1.663	3.728	5.929
combo_bma	0.277	1.520	3.860	6.162
medium_conj	0.400	1.750	3.777	5.951
ucsv	0.357	1.694	3.819	6.028
medium_minn	0.341	1.599	3.722	6.257
small_loose_p5	0.370	1.732	3.932	6.210

unemp_rate

Model	h=1	h=4	h=8	h=12
small_ss	0.337	0.891	1.205	1.223
ar4	0.413	0.915	1.174	1.300
combo_equal	0.369	0.902	1.262	1.321
small_minn	0.349	0.948	1.338	1.366
combo_pool	0.426	0.919	1.260	1.400
small_tight	0.362	0.953	1.351	1.371
small_sv	0.346	0.946	1.395	1.360
medium_minn	0.336	0.918	1.377	1.466
ucsv	0.606	0.974	1.259	1.334
combo_bma	0.339	0.940	1.507	1.416
small_loose_p5	0.354	1.005	1.395	1.470
rw	0.391	0.984	1.391	1.509
combo_logscore	0.583	1.047	1.307	1.385
medium_conj	0.375	1.050	1.487	1.574
ucmean	1.260	1.326	1.435	1.542

cash_rate

Model	h=1	h=4	h=8	h=12
small_ss	0.278	0.990	1.908	2.579
ar4	0.307	1.103	1.931	2.467
combo_equal	0.342	0.969	1.923	2.783
combo_logscore	0.264	0.975	2.141	2.869

Model	h=1	h=4	h=8	h=12
small_minn	0.292	1.045	2.033	2.895
ucsv	0.354	1.190	2.058	2.684
rw	0.352	1.194	2.062	2.684
combo_pool	0.279	1.008	2.161	2.856
combo_bma	0.282	1.049	2.397	2.967
small_tight	0.313	1.190	2.230	3.078
small_sv	0.256	0.978	2.324	3.425
small_loose_p5	0.355	1.130	2.221	3.433
medium_conj	0.479	1.218	2.280	3.418
medium_minn	0.248	1.033	2.525	4.098
ucmean	2.576	2.658	2.649	2.635

2d. Level density calibration – mean log predictive density (higher better)

Averaged across the 4 targets, on the level forecast. The mean log score is brutally sensitive to tail events: $-\infty$ means at least one origin where the realization fell outside that member's predictive support (an individual model can catastrophically miss a tail). The **combinations never do** – the linear pool always assigns positive density – which is the clearest single piece of evidence that pooling buys calibration and robustness.

Model	h=1	h=4	h=8	h=12
combo_equal	-1.852	-1.573	-2.325	-2.751
combo_logscore	-1.840	-1.640	-2.567	-2.929
combo_pool	-1.837	-1.686	-2.536	-2.926
combo_bma	-2.056	-1.974	-3.147	-3.196
medium_minn	-3.127	-2.740	-3.785	-3.335
small_sv	-4.600	-3.141	-4.015	-3.512
ucsv	-5.404	-6.356	-8.782	-7.300
small_loose_p5	-6.527	-6.058	-9.956	-6.763
small_minn	-6.573	-6.665	-11.009	-10.964
ar4	$-\infty$	-4.390	-5.271	-6.282
medium_conj	$-\infty$	-4.830	-10.309	-10.904
rw	$-\infty$	-4.619	-4.431	-4.180
small_ss	$-\infty$	-6.440	-10.932	-8.827
small_tight	$-\infty$	-9.674	-15.438	-14.541
ucmean	$-\infty$	-5.804	-6.761	-10.352

3. Do the combinations beat the best single model?

Mean **level** CRPS over all 4 targets, by horizon bucket. The honest test of a pool is whether it beats both equal weights and the best individual member. (Buckets average level errors of differing scale across horizons, so use this within a bucket to rank models, not to compare buckets.)

Model	near (1-4)	medium (5-8)	far (9-12)
combo_equal	0.533	1.189	1.812
ar4	0.578	1.259	1.849
small_ss	0.554	1.275	1.903

Model	near (1-4)	medium (5-8)	far (9-12)
small_sv	0.534	1.267	1.992
small_minn	0.564	1.294	1.957
medium_minn	0.525	1.264	2.094
small_tight	0.590	1.327	1.994
ucsv	0.590	1.304	2.033
medium_conj	0.588	1.323	2.052
small_loose_p5	0.583	1.364	2.098
combo_pool	0.547	1.359	2.188
combo_logscore	0.610	1.687	2.252
ucmean	1.027	1.576	2.042
combo_bma	0.667	2.191	2.637
rw	0.812	2.063	3.629

4. Statistical significance (Diebold-Mariano)

How often each combination **significantly beats** the random-walk and AR(4) benchmarks on **level** CRPS (Harvey-corrected, 10% level), counted over the 4 targets x 3 horizons {4, 8, 12} tested. A negative DM statistic means the combination is more accurate; significance is one-sided here. Level errors are high-variance (they accumulate the path and are dominated by a few episodes such as 2020), so the DM test has low power on them — the combinations beat the benchmarks on *average* (§3) more often than they do *significantly*.

Combination	beats ar4	beats rw
combo_bma	0 / 12	1 / 12
combo_equal	2 / 12	1 / 12
combo_logscore	1 / 12	1 / 12
combo_pool	0 / 12	1 / 12

Strongest results (significant at 5% or better):

- **combo_equal** beats **ar4** on **unemp_rate** at h=12 (DM -3.37***)
- **combo_equal** beats **ar4** on **cash_rate** at h=4 (DM -2.18**)
- **combo_logscore** beats **ar4** on **unemp_rate** at h=12 (DM -2.10**)
- **combo_logscore** beats **rw** on **gdp_growth** at h=4 (DM -2.05**)

5. Model profiles

One entry per model: its specification, what makes it distinct, the role it plays in the suite, its strengths and failure modes, and where it actually ranks in this evaluation (the eval line is computed, not asserted). Full rationale is in README.md.

5a. VAR members

small_minn

Spec: Independent Normal-inverse-Wishart (Gibbs); 8-variable small SOE core; 4 lags; GLP marginal-likelihood shrinkage; constant volatility; LP scaling (COVID); sum-of-coefficients + dummy-initial-observation priors.

Distinctive: The workhorse Minnesota BVAR, estimated by Gibbs with an *independent* Normal-inverse-Wishart prior — the engine required to impose block exogeneity, since asymmetric (equation-specific) prior variances cannot be represented by a Kronecker/conjugate prior. Shrinkage is data-driven (GLP marginal-likelihood), and sum-of-coefficients + dummy-initial-observation priors discipline the I(1) levels (rates, real TWI).

Role: The central, representative small-SOE BVAR — the reference point the other small members are deliberate variations around (tighter, looser, steady-state, SV).

Strengths & failure modes: Solid all-rounder at short-to-medium horizons. Constant volatility means it leans on the LP scaling for 2020; without it the 2020 outliers would distort the coefficients.

In this evaluation (level error): strongest at cash_rate (medium (5-8)), ranked 2 of 11 individual models.

See: README.md D3, D4, D5, D17

small_ss

Spec: Steady-state (Villani); 8-variable small SOE core; 4 lags; GLP marginal-likelihood shrinkage; constant volatility; LP scaling (COVID).

Distinctive: Reparametrised around its *unconditional means* (Villani steady state), with informative priors placed directly on long-run levels — inflation 2.5% (target midpoint), NAIRU 4.5%, neutral cash rate 3.5%, US potential growth — and on the foreign block's steady states, which the domestic forecast inherits.

Role: The **long-horizon anchor**. An iterated VAR reverts to its unconditional mean at $h = 8-12$; this member makes that mean economically grounded rather than the raw sample average.

Strengths & failure modes: Strongest at medium/far horizons for the mean-reverting targets. Vulnerable if a steady-state anchor is stale (e.g. a shifted neutral rate drags the long end); constant volatility under-disperses around 2020 absent the LP correction.

In this evaluation (level error): best individual model for unemp_rate at medium (5-8); unemp_rate at far (9-12); cash_rate at medium (5-8).

See: README.md D3, D4, D5, D17

small_sv

Spec: Stochastic volatility, equation-by-equation; 8-variable small SOE core; 4 lags; GLP marginal-likelihood shrinkage; stochastic volatility; t-errors (SV-t) (COVID).

Distinctive: Stochastic volatility estimated equation-by-equation (the Carriero-Clark-Marcellino triangular factorisation), with t-distributed errors (the CCMM SV-t COVID treatment). Block exogeneity is *exact* — foreign equations simply drop the domestic regressors.

Role: The **density-calibration specialist**: time-varying volatility tracks changing macro uncertainty and the fat tails absorb outliers instead of letting them widen the whole history.

Strengths & failure modes: Its SV captures the time-varying conditional variance, so it stays well-calibrated when uncertainty shifts — competitive on the GDP level at medium horizons. The volatility state at the jump-off can over/under-shoot if the last few quarters were unusual; the small system limits cross-variable information.

In this evaluation (level error): best individual model for gdp_growth at medium (5-8); gdp_growth at far (9-12); cash_rate at near (1-4).

See: README.md D3, D6, D17

small_loose_p5

Spec: Independent Normal-inverse-Wishart (Gibbs); 8-variable small SOE core; 5 lags; fixed shrinkage $\lambda=0.4$; constant volatility; LP scaling (COVID).

Distinctive: A deliberately *under-shrunk*, longer-lag variant — fixed $\lambda = 0.4$ (vs the $\sim 0.1-0.15$ the GLP procedure selects) and 5 lags — so the data speak more and richer dynamics can show through, at the cost of estimation noise.

Role: The **loose / long-lag** diversity axis: it fails differently from the tightly-shrunk members and can capture dynamics they shrink away.

Strengths & failure modes: Occasionally best at near-horizon density when the extra flexibility pays; noisier and prone to wider intervals at long horizons (the cost of light shrinkage in a short sample).

In this evaluation (level error): strongest at gdp_growth (near (1-4)), ranked 4 of 11 individual models.

See: README.md D4, D8

medium_minn

Spec: Stochastic volatility, equation-by-equation; 13-variable medium system; 2 lags; GLP marginal-likelihood shrinkage; stochastic volatility; t-errors (SV-t) (COVID).

Distinctive: The larger system — 13 variables (adds terms of trade, wages, employment, consumption, the 10y yield) with stochastic volatility and t-errors, but only 2 lags to keep the parameter count feasible; equation-by-equation estimation keeps the recursive loop tractable.

Role: The **medium-system** axis (Banbura-Giannone-Reichlin): medium systems often forecast best given enough shrinkage, and the extra variables bring cross-sectional information the small core lacks.

Strengths & failure modes: Strong short-horizon density (it tends to win the near bucket). The short lag length limits long-horizon dynamics, and more parameters mean more estimation uncertainty at the far end.

In this evaluation (level error): best individual model for gdp_growth at near (1-4); unemp_rate at near (1-4).

See: README.md D1, D6, D17

medium_conj

Spec: Block-recursive conjugate NIW; 13-variable medium system; 4 lags; fixed shrinkage $\lambda=0.1$; constant volatility; LP scaling (COVID); sum-of-coefficients + dummy-initial-observation priors.

Distinctive: The medium system estimated by the fast *block-recursive conjugate* scheme (foreign VAR + domestic block conditioned on contemporaneous foreign values, the RBNZ Bloor-Matheson approach) — closed-form, so cheap even at 13 variables x 4 lags. Block exogeneity is exact by the recursive structure, not the prior.

Role: The cheap medium workhorse; it complements `medium_minn` (conjugate constant-volatility vs SV) on the same large system.

Strengths & failure modes: Tends to lead the far-horizon GDP level error. Constant volatility leans on LP for 2020; the conjugate Kronecker prior cannot represent asymmetric shrinkage, which is why block exogeneity comes from the recursive structure.

In this evaluation (level error): strongest at gdp_growth (far (9-12)), ranked 4 of 11 individual models.

See: README.md D3, D5, D8

small_tight

Spec: Block-recursive conjugate NIW; 8-variable small SOE core; 4 lags; fixed shrinkage $\lambda=0.05$; constant volatility; LP scaling (COVID); sum-of-coefficients + dummy-initial-observation priors.

Distinctive: The heavily-shrunk small model — fixed $\lambda = 0.05$, far tighter than the GLP selection — pulling hard toward the persistence/random-

walk prior, so it is parsimonious and low-variance.

Role: The **tight** diversity axis and the long-horizon robustness member: heavy shrinkage buys stability where lightly-parametrised models wander.
Strengths & failure modes: Best or near-best at far-horizon GDP (the tight prior stops it over-reacting). Can be too rigid at short horizons, missing genuine dynamics the looser members catch.

In this evaluation (level error): strongest at gdp_growth (far (9-12)), ranked 2 of 11 individual models.

See: README.md D4, D5, D8

5b. Benchmark members

rw

Spec: Random walk; LP scaling (COVID).

Distinctive: The no-change forecast: the last observed value persists, with Gaussian increments scaled to the historical change.

Role: The universal hard-to-beat short-horizon bar for persistent/level variables, and a pool member.

Strengths & failure modes: Competitive at $h = 1$ for level variables; fails badly at long horizons — its level path runs away, which the level-error tables (§2) expose brutally.

In this evaluation (level error): best individual model for cpi_inflation at near (1-4); cpi_inflation at medium (5-8).

See: README.md D8

ar4

Spec: Bayesian AR(4); LP scaling (COVID).

Distinctive: A Bayesian AR(4) per variable with Minnesota-style lag shrinkage and a stationarity-truncated posterior.

Role: The univariate-persistence bar — it isolates how much of the forecast is just own-history dynamics.

Strengths & failure modes: Surprisingly strong for inflation at near/medium horizons, where univariate dynamics dominate; it cannot use cross-variable information, so it lags when that matters.

In this evaluation (level error): best individual model for cpi_inflation at far (9-12); cash_rate at far (9-12).

See: README.md D8

ucsv

Spec: Unobserved components + stochastic volatility; t-errors (robust) (COVID).

Distinctive: Stock-Watson unobserved-components stochastic volatility per variable: a random-walk trend plus transitory noise, both with time-varying variances and outlier-robust t-errors.

Role: The canonical inflation benchmark and a genuine density anchor for the other targets.

Strengths & failure modes: Strong for inflation (its native use case); weaker for variables with richer multivariate dynamics. The trend/noise split is weakly identified, so it is gated on Monte-Carlo precision, not raw ESS.

In this evaluation (level error): strongest at gdp_growth (near (1-4)), ranked 3 of 11 individual models.

See: README.md D9, D17

ucmean

Spec: Unconditional mean; LP scaling (COVID).

Distinctive: A Gaussian density centred on the expanding-sample mean with the sample variance — the simplest possible density forecast.

Role: The floor: the ‘did the model beat just predicting the long-run average’ bar.

Strengths & failure modes: Unexpectedly competitive at long horizons for mean-reverting growth (everything reverts to the mean eventually); useless at short horizons where dynamics matter.

In this evaluation (level error): strongest at cpi_inflation (far (9-12)), ranked 3 of 11 individual models.

See: README.md D8

5c. Combination schemes

combo_equal

Weights: Equal weights on every member, per variable x horizon bucket.

In this evaluation: The forecast-combination-puzzle benchmark and the recommended robust default: in the evaluation it has the best mean log score at every horizon and the best far-horizon CRPS. Hard to beat because it never over-fits weights.

combo_logscore

Weights: Weights proportional to each member’s recent log predictive score, with a forgetting factor, shrunk toward equal.

In this evaluation: Adapts to which members are forecasting well lately; the shrinkage and forgetting guard against over-concentrating on a member that was lucky.

combo_pool

Weights: Optimal prediction pool (Hall-Mitchell / Geweke-Amisano): weights on the simplex that maximise the historical *pooled* log score, shrunk toward equal.

In this evaluation: Unlike BMA it does not degenerate to a single model (‘all models are false but useful’); competitive with equal weights and occasionally better at near horizons.

combo_bma

Weights: Bayesian model averaging by predictive likelihood — no shrinkage.

*In this evaluation: **Diagnostic only.*** It concentrates weight on the single best-fitting member, so it answers ‘which model does the data favour’ rather than serving as a robust combination; reported, not recommended.

6. How to read this

- **The metric is the level error.** For GDP and inflation it is the cumulative level from the forecast origin (where real GDP and the price level land h quarters out); for unemployment and the cash rate it is the rate level at $t+h$. Level errors accumulate the whole forecast path, so they grow with horizon and are far more discriminating than the single-quarter growth rate — a model that gets the persistent/drift component wrong (e.g. the random walk) is exposed here. The quarterly and year-ended *growth* scores live in `output/tables/scores_by_horizon.csv` if you need them.
- **Point gains over the best member are modest by design** — equal weights are hard to beat (the forecast-combination puzzle). The pool’s payoff is *calibration and robustness*: it insures against any single member failing, rather than always winning on accuracy.
- **CRPS and log score can disagree** on outlier-heavy windows (log score is far more sensitive to tail events); both are reported. A COVID-excluded variant is in `output/tables/scores_by_horizon_excovid.csv`. See the full Quarto report for fan charts, PIT calibration, and weight-evolution plots.